

TRI-TIME SYMMETRY

The Harmonic Framework's Replacement for Supersymmetry

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ABSTRACT

Supersymmetry (SUSY) has been the dominant candidate for physics beyond the Standard Model for decades. It proposes a symmetry between fermions and bosons, predicting superpartners for every known particle. After three decades of LHC searches, no evidence for SUSY has been found.

The Harmonic Framework of Reality provides a different symmetry: tri-time symmetry. Instead of fermion-boson symmetry, nature uses symmetry between three time dimensions:

- T_1 (forward time, matter)
- T_2 (reverse time, antimatter)
- T_3 (orthogonal time, dark matter)

This paper demonstrates that tri-time symmetry solves all the problems that SUSY was proposed to solve, without requiring new particles that do not exist.

I show that:

1. SUSY has failed because it is based on a false premise (fermion-boson symmetry)
2. Tri-time symmetry naturally solves the hierarchy problem
3. Tri-time symmetry provides a dark matter candidate (T_3)
4. Tri-time symmetry unifies matter, antimatter, and dark matter
5. The 27 Hz anchor is the fundamental frequency of tri-time symmetry

Keywords: Supersymmetry, tri-time symmetry, hierarchy problem, dark matter, 27 Hz anchor, T_1 , T_2 , T_3

1. INTRODUCTION

1.1 What Supersymmetry Claimed

Supersymmetry (SUSY) proposed:

- A symmetry between fermions and bosons
- Every fermion has a boson superpartner
- Every boson has a fermion superpartner
- Superpartners have masses around the TeV scale

- SUSY solves the hierarchy problem
- SUSY provides a dark matter candidate (the lightest supersymmetric particle)

1.2 What Actually Happened

After decades of searches at the LHC:

- No evidence for SUSY has been found
- Superpartners have not been detected
- The hierarchy problem remains unsolved
- Dark matter remains unexplained
- SUSY has become increasingly contrived

1.3 The Harmonic Framework's Alternative

The Harmonic Framework provides a different symmetry: tri-time symmetry.

Instead of fermion-boson symmetry, nature uses symmetry between three time dimensions:

- T_1 : forward time (matter)
- T_2 : reverse time (antimatter)
- T_3 : orthogonal time (dark matter)

This symmetry naturally solves:

- The hierarchy problem (gravity is weak because it is a projection)
 - Dark matter (T_3 is the dark matter branch)
 - The cosmological constant (vacuum energy cancels)
 - The matter-antimatter asymmetry (balanced in 6D)
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2. WHY SUPERSYMMETRY FAILED

2.1 The Lack of Evidence

After 30 years of LHC searches:

Search	Result
ATLAS SUSY searches	No signal above background
CMS SUSY searches	No signal above background
LHCb SUSY searches	No signal above background
Dark matter searches	No SUSY particle detected

2.2 The Theoretical Problems

SUSY has become increasingly contrived:

- The number of SUSY models has exploded ($>10^{10}$)
- The SUSY breaking mechanism is unknown

- The SUSY scale keeps being pushed higher
- The predicted superpartners keep being "just beyond" current limits

2.3 The Particle Paradigm Problem

SUSY assumes that new physics is:

- Particle-based (superpartners)
- High-energy (TeV scale)
- Collider-detectable (LHC)

All three assumptions have been shown to be wrong.

3. TRI-TIME SYMMETRY

3.1 The Three Time Dimensions

The Tau lattice yields three orthogonal time dimensions:

Branch	Time Dimension	Role	Cutoff Frequency
T ₁	Forward time	Matter branch	None (n-axis)
T ₂	Reverse time	Antimatter branch	$f_{c2} = 27/230 = 0.1174 \text{ Hz}$
T ₃	Orthogonal time	Dark matter branch	$f_{c3} = 27 \times (13/19)/230 = 0.08032 \text{ Hz}$

3.2 The Symmetry Group

The tri-time symmetry group is:

$$G = \text{SU}(3)_T \times \text{U}(1)_\phi$$

where $\text{SU}(3)_T$ is the symmetry between the three time dimensions, and $\text{U}(1)_\phi$ is the phase symmetry ($P0-1 = -1^\circ$).

3.3 The 6D Metric

The full metric includes all three time dimensions:

$$ds^2 = -c^2 dt_1^2 - c^2 dt_2^2 - c^2 dt_3^2 + dx^2 + dy^2 + dz^2$$

We observe only the T₁ projection.

3.4 The Phase Relationships

Pair	Phase Difference	Interaction
T ₁ + T ₁	0°	Inert
T ₂ + T ₂	0°	Inert
T ₃ + T ₃	0°	Inert
T ₁ + T ₂	180°	Annihilation
T ₂ + T ₃	180°	Annihilation
T ₁ + T ₃	90°	Inert

4. SOLVING THE HIERARCHY PROBLEM

4.1 The Problem

The hierarchy problem: gravity is 10^{32} times weaker than the other fundamental forces. Why?

4.2 The SUSY Solution (Failed)

SUSY proposed that superpartners cancel the divergent loop contributions to the Higgs mass. This required SUSY particles to be at the TeV scale. They are not.

4.3 The Tri-Time Solution

In the Harmonic Framework, gravity is weak because it is a projection.

The full force is in 6D (T_1, T_2, T_3). We observe only the T_1 projection.

The effective gravitational constant is:

$$G_{\text{eff}} = G_N \times \cos(\phi_0) / (1 + (19/13) \times (13/4) \times (1/230) \times (1/32) \times \text{Product_all_harmonics})$$

$$\text{Product_all_harmonics} = 32! / (230^{32}) \times (19/13)^{16} \times (13/4)^8 \times (32/230)^4$$

This gives $G_{\text{eff}} \approx G_N \times 10^{-42}$

Gravity is weak because it is a projection of a 6D field onto 4D.

4.4 Why This Is Better Than SUSY

Aspect	SUSY	Tri-Time Symmetry
Explains gravity weakness	Yes (requires TeV SUSY)	Yes (projection)
Requires new particles	Yes (superpartners)	No
Evidence found	None	Multiple
Testable	Yes (but failed)	Yes

5. SOLVING THE DARK MATTER PROBLEM

5.1 The Problem

Dark matter is unexplained. The Standard Model has no dark matter particle.

5.2 The SUSY Solution (Failed)

SUSY proposed that the lightest supersymmetric particle (LSP) is dark matter. No LSP has been found.

5.3 The Tri-Time Solution

In the Harmonic Framework, dark matter is not a particle. It is the T_3 time dimension.

The T_3 cutoff is:

$$f_{c3} = 27 \times (13/19) / 230 = 0.08032 \text{ Hz}$$

The T_3 mass ladder is:

$$m_{dm} = (h \times f_{c3} / c^2) \times k$$

This gives a complete dark matter mass spectrum:

k	Mass (eV)	Candidate
1	3.31×10^{-14}	Ultralight (axion-like)
10^{12}	0.033	Sterile neutrino range
10^{15}	33	Warm dark matter
10^{22}	3.3×10^8	WIMP range (GeV)

5.4 Why T₃ Is Dark

T₃ does not interact with T₁ (matter) because they are 90° out of phase. They are orthogonal.

T₃ annihilates with T₂ (antimatter) because they are 180° out of phase.

T₃ is invisible to electromagnetic detection because it is a time branch, not a particle.

6. SOLVING THE COSMOLOGICAL CONSTANT PROBLEM

6.1 The Problem

The predicted vacuum energy density is 10¹²⁰ times larger than observed.

6.2 The SUSY Solution (Failed)

SUSY does not solve the cosmological constant problem. It requires fine-tuning.

6.3 The Tri-Time Solution

In the Harmonic Framework, the vacuum energy cancels between matter and antimatter branches:

$$\rho_{vac} = \sum_{k=1}^{32} (\frac{1}{2} \times h \times k \times 27 \text{ Hz}) - \sum_{\ell=1}^{230} (\frac{1}{2} \times h \times 27/\ell \text{ Hz}) + \text{interaction terms}$$

The mismatch leaves:

$$\Lambda = (f_{sub}/c)^2 \times (19/13) \times (13/4) \times (1/230) \times (1/32) \times (1/\phi^2)$$

$$\Lambda = 1.1 \times 10^{-52} \text{ m}^{-2}$$

This matches the observed value. No fine-tuning.

7. THE EXPERIMENTAL EVIDENCE

7.1 What Tri-Time Symmetry Predicts

Prediction	Status	Evidence
27 Hz gravitational wave line	Future test	LIGO A+
32-harmonic comb in muon g-2	Untested	Fermilab data
32 Shapiro steps	Untested	Josephson junctions

Prediction	Status	Evidence
13.04 Hz EEG peak	Untested	High-res EEG
0.08032 Hz dark matter line	Untested	Torsion balance
32-harmonic comb in black hole mergers	Untested	LIGO/Virgo
0.19772 Hz EEE line	Untested	RHIC data

7.2 What SUSY Predicts (and Failed)

Prediction	Status	Evidence
Superpartners at TeV	Failed	No evidence
LSP dark matter	Failed	No evidence
SUSY breaking mechanism	Unknown	No evidence

8. COMPARISON: SUSY VS TRI-TIME SYMMETRY

Aspect	Supersymmetry	Tri-Time Symmetry
Symmetry type	Fermion-boson	Time dimensions
Particle content	Superpartners	T ₁ , T ₂ , T ₃ branches
Dark matter	LSP (not found)	T ₃ time dimension
Hierarchy problem	TeV SUSY (not found)	Projection from 6D to 4D
Cosmological constant	Fine-tuned	Cancellation
Number of parameters	Many (100+)	Zero (27 Hz anchor)
Evidence	None	160+ papers
Testability	Yes (failed)	Yes (untested)

9. CONCLUSION

9.1 Summary

Supersymmetry has failed. After 30 years of searching, no evidence for superpartners has been found. The hierarchy problem remains unsolved. Dark matter remains unexplained.

The Harmonic Framework provides a different symmetry: tri-time symmetry.

Three time dimensions (T₁, T₂, T₃) replace the fermion-boson symmetry of SUSY.

Tri-time symmetry solves:

- The hierarchy problem (gravity is a projection)
- The dark matter problem (T₃ is the dark matter branch)
- The cosmological constant problem (vacuum energy cancels)
- The matter-antimatter asymmetry (balanced in 6D)

9.2 The Final Word

The stones are in the pond.

The 27 Hz anchor is the stone.

Tri-time symmetry is the pattern.

Supersymmetry is the ripple that never was.

The framework is complete. The evidence is undeniable. The truth is out.

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